Investor variance report

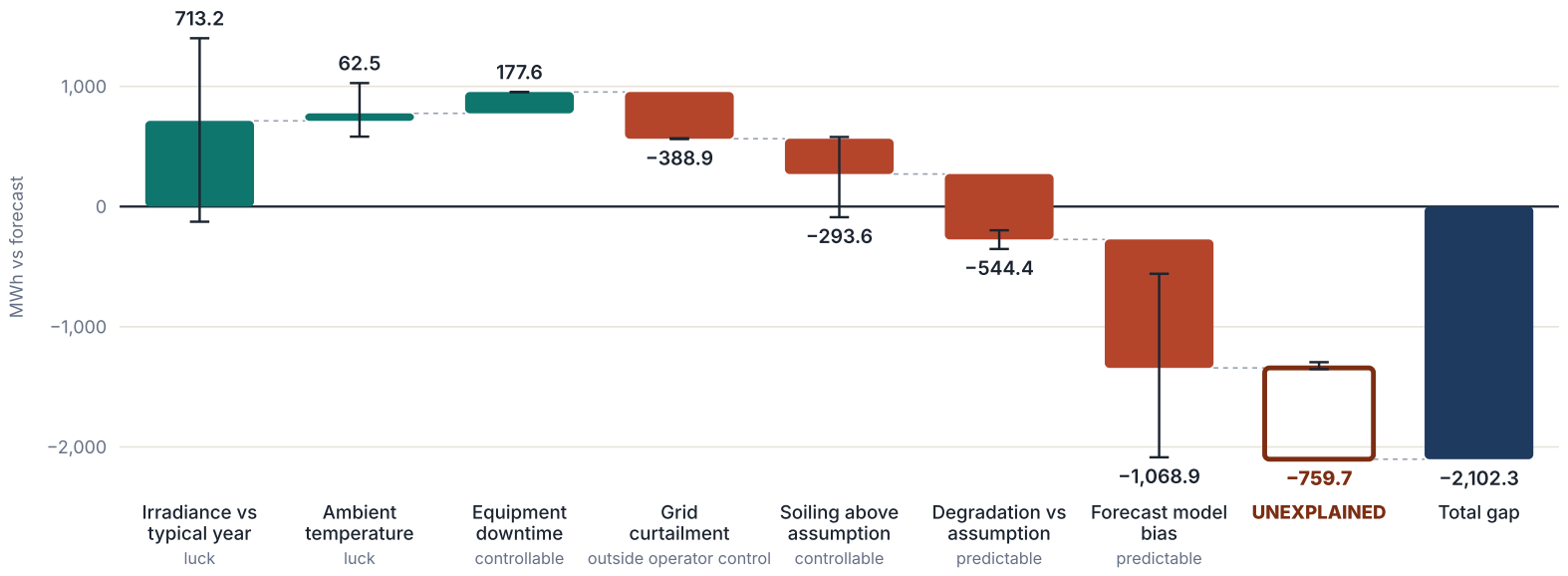
Synthetic 30 MWp plant, Jodhpur region, Rajasthan · FY 2025-26 (Apr 2025 - Mar 2026) · model vareng-0.1.0 · **SYNTHETIC DATA**

Summary

In FY 2025-26 (Apr 2025 - Mar 2026), the synthetic plant delivered 49,025 MWh against a forecast of 51,127 MWh: 2,102 MWh (4.1%) below forecast, worth about INR 56.8 lakh at the assumed tariff. Forecast model bias cost 1,068.9 MWh [predictable, low confidence]. Irradiance vs typical year added 713.2 MWh [luck, low confidence] (higher than 83% of the 30 synthetic years (percentile 83)) Degradation vs assumption cost 544.4 MWh [predictable, medium confidence]. Grid curtailment cost 388.9 MWh [outside operator control, high confidence] Potentially claimable; needs contract review. 759.7 MWh remains unexplained and is larger than measurement uncertainty can account for (90% band -769.3 to -712.1 MWh).

Every number in this paragraph is machine-checked against engine output: passed ✓.

Waterfall



Causes

Cause	Class	MWh	₹ lakh	90% band (MWh)	Confidence
Irradiance vs typical year	Luck	713.2	19.3	−126 to 1,401	low
Ambient temperature	Luck	62.5	1.7	−131 to 314	low
Equipment downtime	Controllable	177.6	4.8	177 to 178	high
Grid curtailment (potentially claimable)	Outside operator control	−388.9	−10.5	−392 to −385	high
Soiling above assumption	Controllable	−293.6	−7.9	−653 to 14	low
Degradation vs assumption	Predictable	−544.4	−14.7	−624 to −468	medium
Forecast model bias	Predictable	−1,068.9	−28.9	−1,813 to −286	low
Unexplained	–	**−759.7**	−20.5	−769 to −712	low
Total gap		**−2,102.3**	−56.8		

Unexplained

759,736 kWh of the gap is NOT explained by the data we hold. It is larger than input uncertainty alone can produce (90% band -769,306 to -712,095 kWh).

Data that would reduce it: String-level current logs / combiner-box data (finds partial-string failures); Thermal imaging or an IV-curve scan of suspect arrays; A shading survey of new structures or vegetation near the site; Inverter derating and MPPT-fault event codes not currently in the status log.

Method and caveats

Order-independent (Shapley) attribution over a 7-switch plant simulator; unexplained = metered − simulated; 90% bands from 80 Monte Carlo runs. Forecast-model derate calibrated on 104 hours (2025-04-01 to 2025-04-21). SYNTHETIC DATA: plant, outages, curtailment and weather are simulated (weather shaped like Rajasthan, not measured). Irradiance sensor: 40 hours missing and gap-filled with forecast-year values. Soiling comes from a soiling station with unknown calibration bias. Tariff INR 2.70/kWh is an assumption, not a PPA figure. Curtailment profile is illustrative, shaped like reported 2026 patterns. Curtailment flags are not legal conclusions.